Harshit Kumar Sharma
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Internships

Project Engineering Intern - DIAT (DU), DRDO, Optical Communication and Bio photonics Lab : LINK **Pune, India**
Underwater Wireless Optical Communication (UWOC) June 2025 – July 2025

- UWOC System Development: Built laser-based optical communication achieving more than 85% classification accuracy in turbid environments using 50K+ NTU-labeled sensor data.
- Tkinter-based GUI: Developed real-time GUI for live data visualization and turbidity prediction on Raspberry pi.
- BER Optimization: Reduced BER by 10% using dynamic threshold tuning and analog signal optimization.

Volunteer Teacher Assistant - Durga Saptashati Foundation (NGO) : LINK **Dwarka, New Delhi**
Electronics & Programming Mentor June 2024 – July 2024

- Mentorship: Guided and taught 20+ students in basic electronics and programming.
- Hands-on Projects: Designed 5 practical electronics projects (LED circuits, etc.) to teach fundamentals.

Projects

Low-Cost Tamper detection in weighing Scale Sep 2025 - Jan 2026
Embedded Systems, IoT, OTA Update, Machine Learning, Android App.

- Developed a low-cost embedded tamper detection system for digital weighing scales to prevent unauthorized manipulation.
- Integrated HX711 load cell amplifier for precise weight measurement and real-time data acquisition.
- Applied unsupervised machine learning - based classification to detect abnormal weight patterns indicating tampering.
- Enabled IoT connectivity for remote monitoring and tamper alerts.
- Implemented OTA firmware updates for remote maintenance and upgrades.
- Developed an Android app (APK) for real-time weight display and alert notifications.
- GitHub Repository Link: https://github.com/harshit2387/Smart-Weighing-machine

Smart AI Desktop Bot Apr 2025 - June 2025
Bluetooth, esp32, Machine Learning Audio Processing.

- Designed and implemented a Bluetooth A2DP Sink on ESP32 to receive wireless audio and stream it via I2S to an external audio amplifier (MAX98357A).
- Configured ESP32 I2S interface using default pin mapping for reliable digital audio transmission.
- Integrated Audio Tools library for efficient Bluetooth audio handling and real-time streaming.
- Worked with INMP114 digital microphone and OLED display for system interaction and status visualization.
- Configured ESP32 partition scheme (OTA: 2MB APP / 2MB SPIFFS) to ensure stable Bluetooth operation.
- Designed, soldered, and tested a custom PCB for embedded audio hardware integration.
- GitHub Repository Link: https://github.com/harshit2387/robo

Certificate

Machine Learning and Deep Learning - NPTEL: **LINK** July 2025 – Oct 2025

Technical Skills

Languages C, C++, Java, Embedded C, Python, Kotlin.
Technologies/Frameworks: Git, GitHub, Microcontrollers (Arduino/ Esp32/ STM32/ Raspberry Pi Pico), Machine Learning, Linux, IoT, Embedded Systems, Blender, Android Studio, FPGA, Excel.
Soft Skills: Teamwork, Adaptability, Attention to Detail, Resilience & Persistence.

Education

Lovely Professional University **2023 – 2027**
Computer Science and Engineering — CGPA: 7.32 Phagwara Punjab
Kendriya Vidyalaya Dwarka Sec - 5 **2022 – 2023**
12th with Science — Percentage: 92.00% Dwarka, New Delhi
Kendriya Vidyalaya Dwarka Sec - 5 **2020 – 2021**
10th with Science — Percentage: 86.25% Dwarka, New Delhi